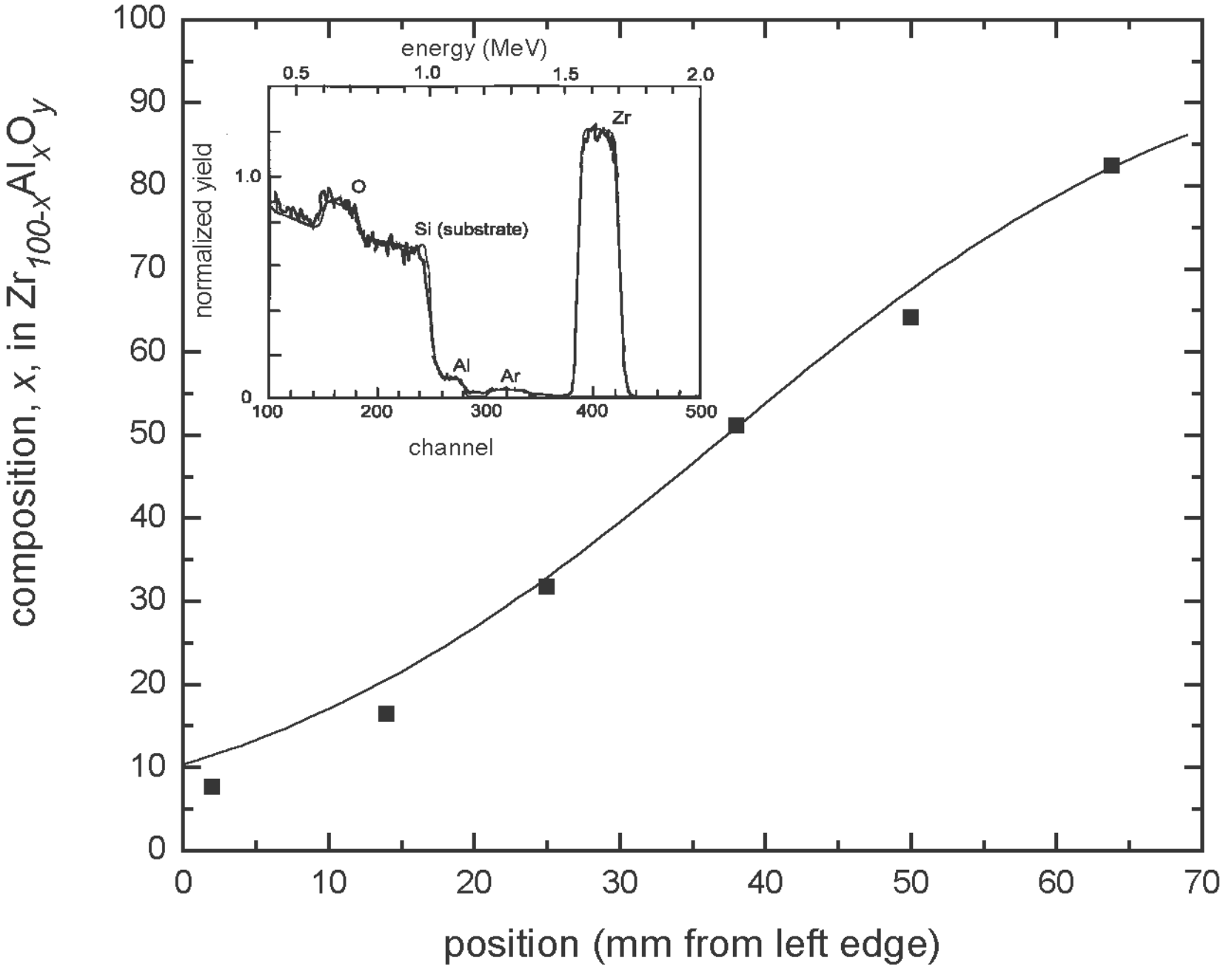
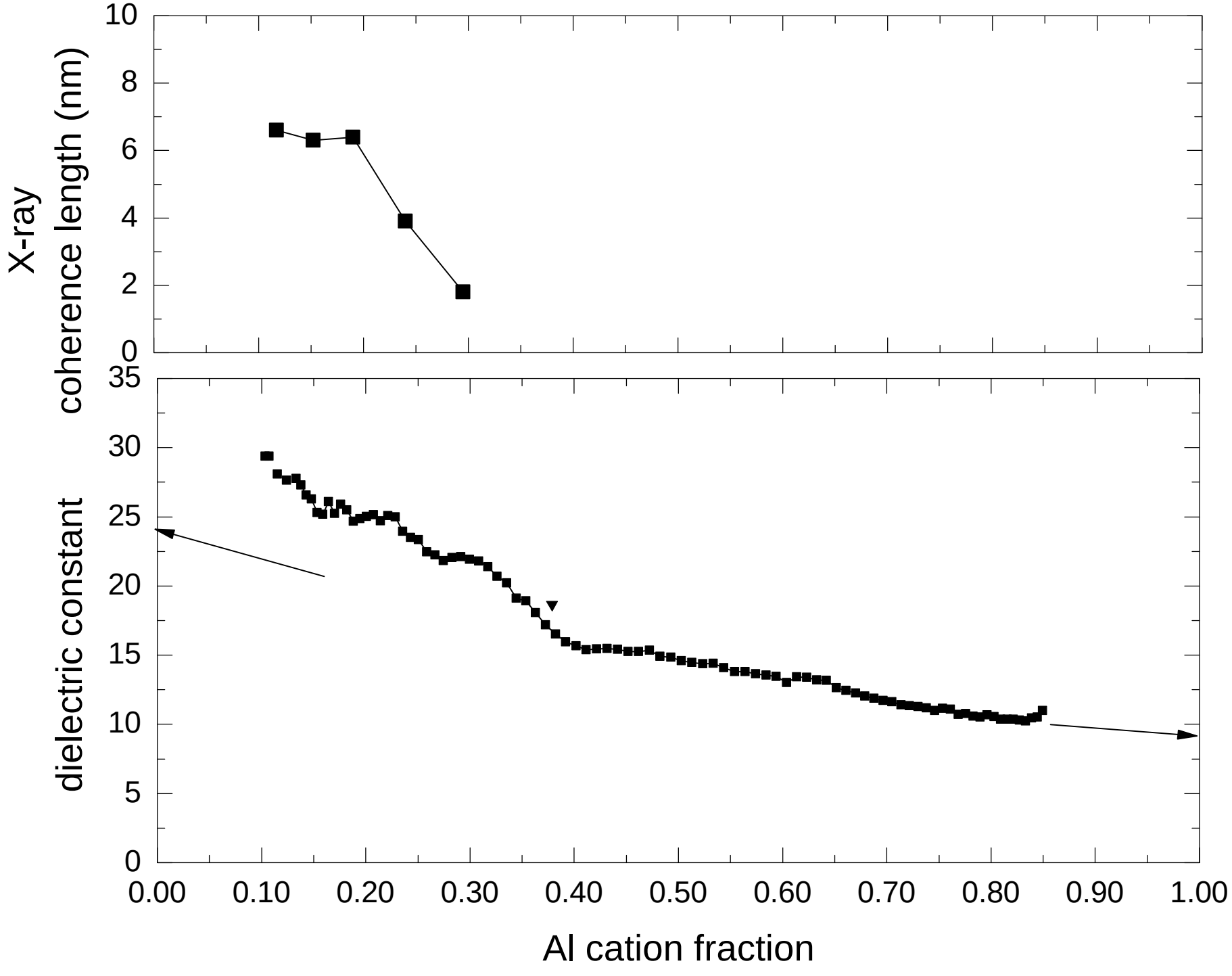
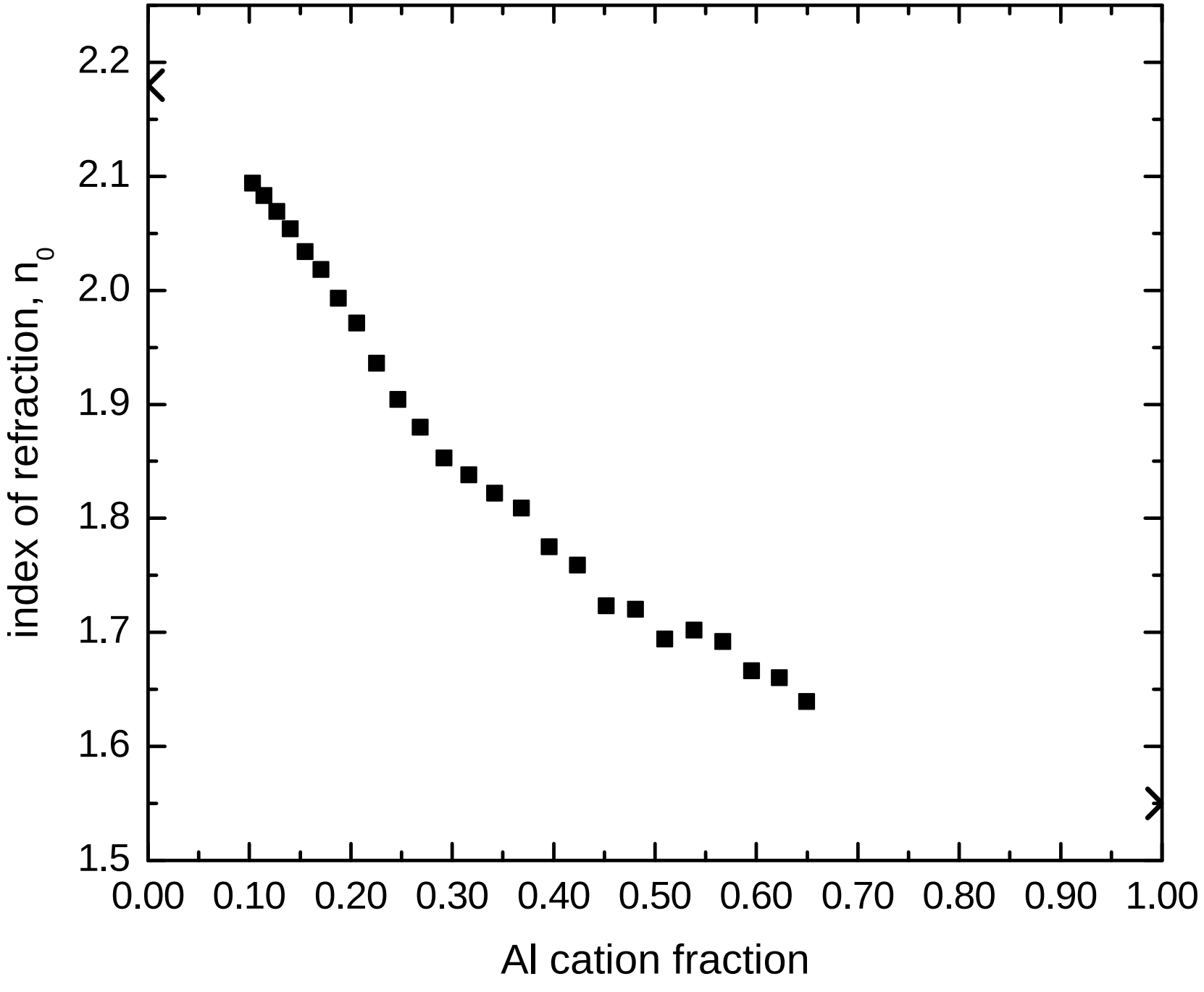
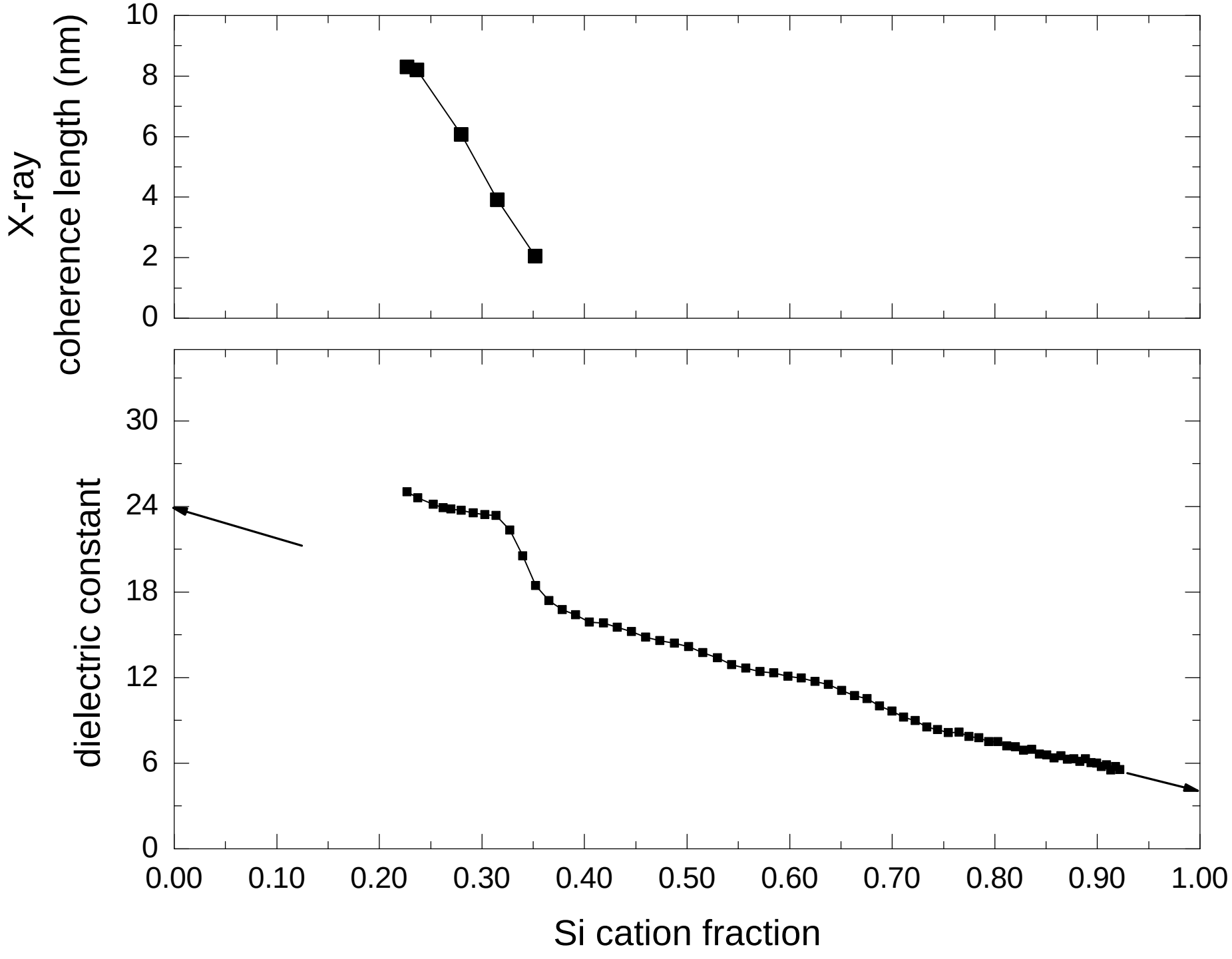
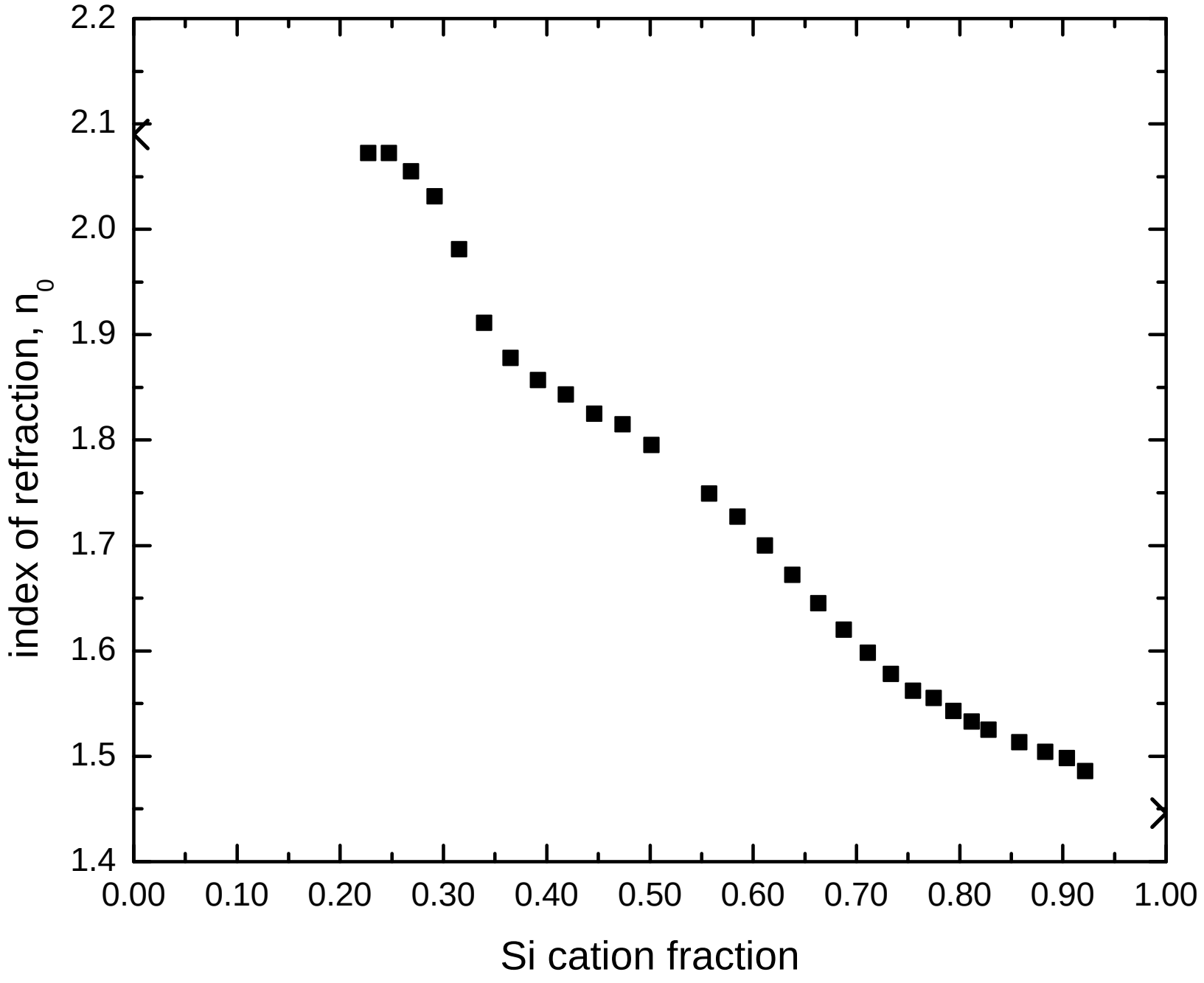
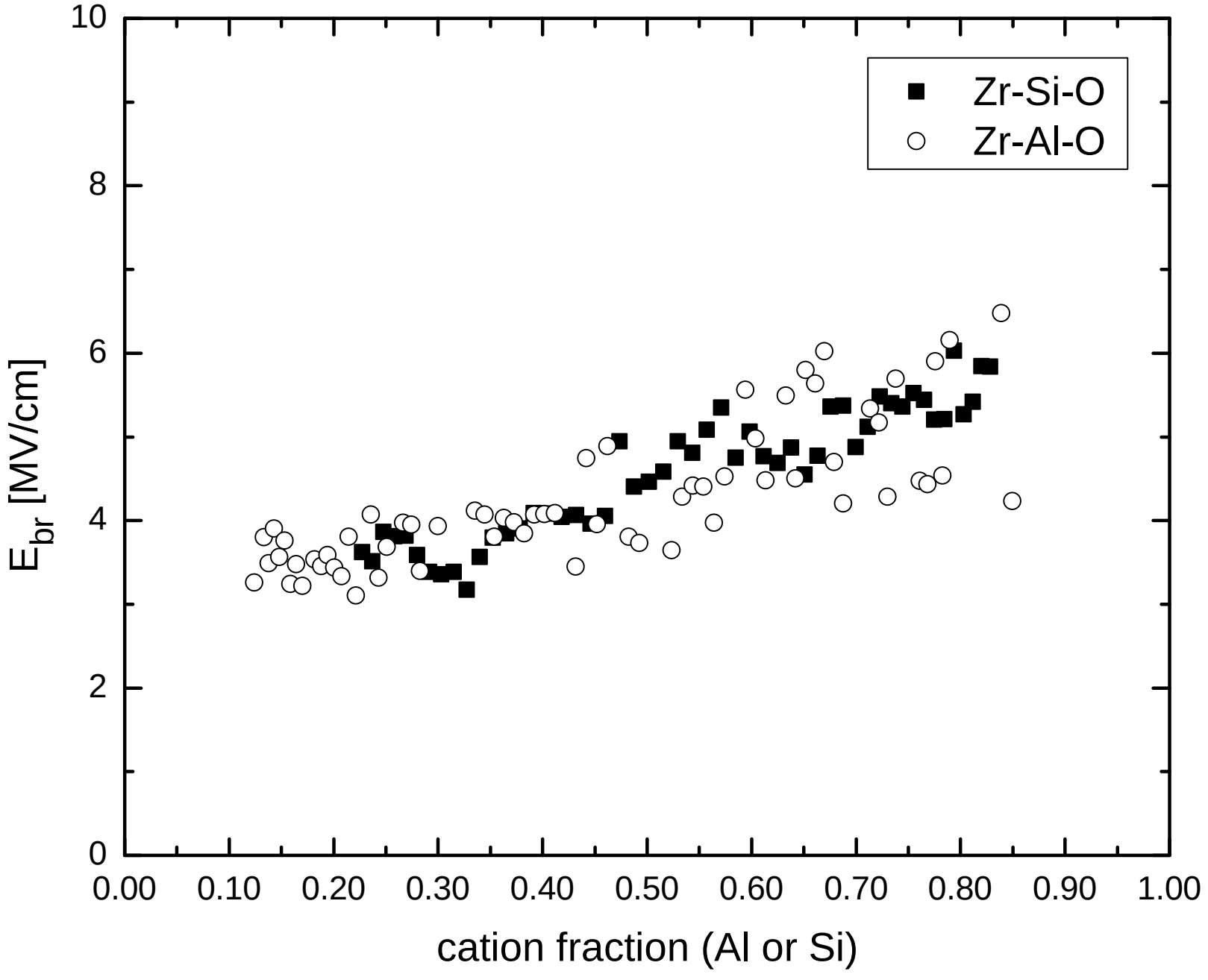

composition, x, in $Zr_{100-x}Al_xO_y$
position (mm from left edge)
0
10
20
30
40
50
60
70
80
90
100
energy (MeV)
0.5
1.0
1.5
2.0
normalized yield
1.0
0
O
Si (substrate)
Al
Ar
Zr
100
200
300
400
500
channel

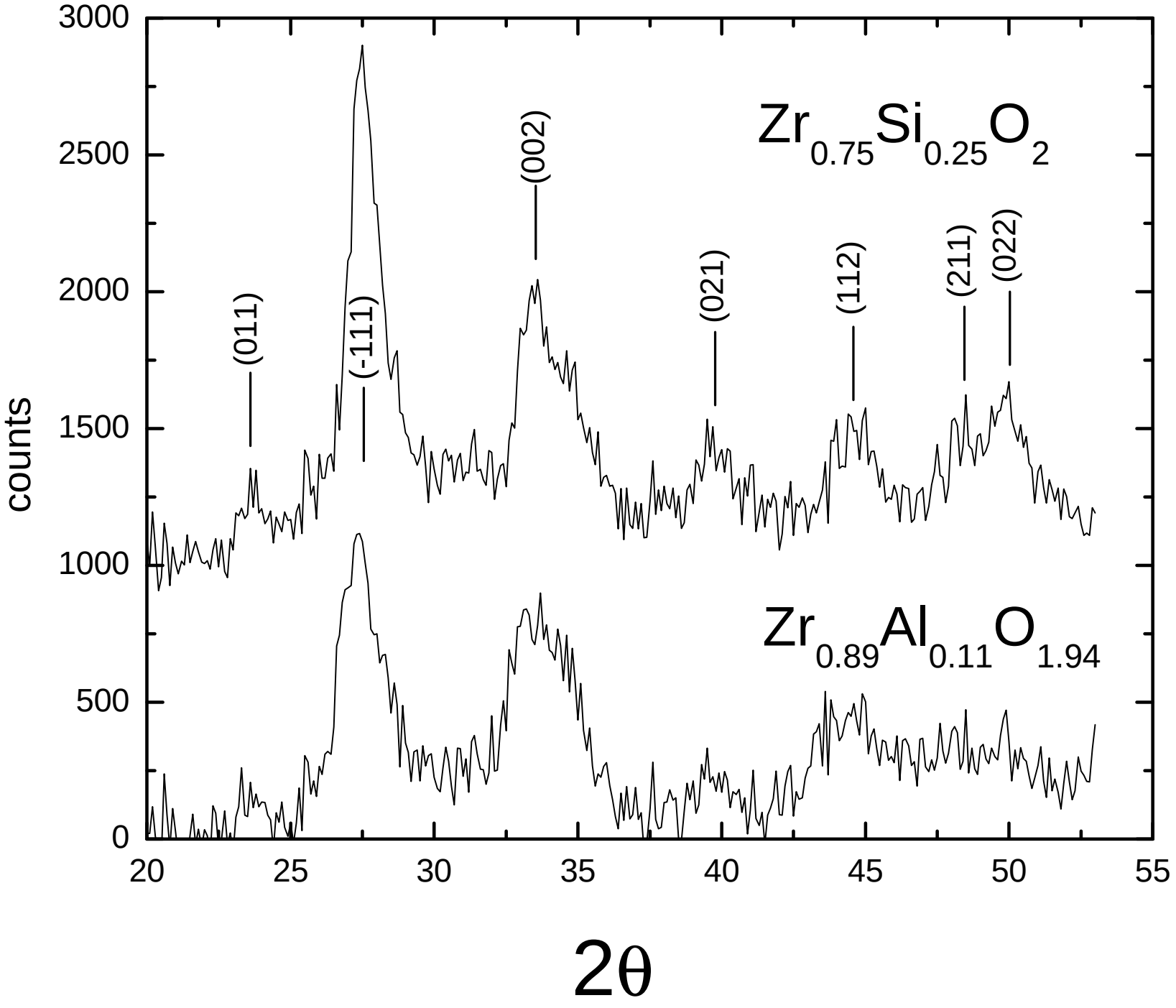

X-ray coherence length (nm)
dielectric constant
Al cation fraction
0
2
4
6
8
10
0
5
10
15
20
25
30
35
0.00
0.10
0.20
0.30
0.40
0.50
0.60
0.70
0.80
0.90
1.00


index of refraction, n_0
2.2
2.1
2.0
1.9
1.8
1.7
1.6
1.5
0.00
0.10
0.20
0.30
0.40
0.50
0.60
0.70
0.80
0.90
1.00
Al cation fraction


X-ray
coherence length (nm)
10
8
6
4
2
0
dielectric constant
30
24
18
12
6
0
0.00
0.10
0.20
0.30
0.40
0.50
0.60
0.70
0.80
0.90
1.00
Si cation fraction


2.2
2.1
2.0
1.9
1.8
1.7
1.6
1.5
1.4
index of refraction, n0
0.00
0.10
0.20
0.30
0.40
0.50
0.60
0.70
0.80
0.90
1.00
Si cation fraction


Zr-Si-O
Zr-Al-O
E_{br} [MV/cm]
cation fraction (Al or Si)
0
2
4
6
8
10
0.00
0.10
0.20
0.30
0.40
0.50
0.60
0.70
0.80
0.90
1.00


3000
2500
2000
1500
1000
500
0
counts
20
25
30
35
40
45
50
55
2θ
(011)
(-111)
(002)
(021)
(112)
(211)
(022)
Zr0.75Si0.25O2
Zr0.89Al0.11O1.94